Module-1

1. C = Set of all complex numbers (Q)

$(a_1+ib_1)\,\rho\,(a_2+ib_2) \iff a_1 \le a_2$ and $b_1 \le b_2$

Prove that (C,ρ) is a Partially Ordered Set (POSET)

We have to show that ρ is –
1. Reflexive
2. Antisymmetric
3. Transitive.

① Reflexive: Let, $z = a+ib \in C$

Since $a \le a \wedge b \le b$

$\therefore (a+ib)\,\rho\,(a+ib) \rightarrow \therefore \rho$ is Reflexive.

② Antisymmetric:

Assume, $(a_1+ib_1)\,\rho\,(a_2+ib_2)$, $a_1 \le a_2 \wedge b_1 \le b_2$

& $(a_2+ib_2)\,\rho\,(a_1+ib_1)$, $a_2 \le a_1 \wedge b_2 \le b_1$

$\therefore a_1 = a_2 \wedge b_1 = b_2$

$\therefore (a_1+ib_1) = (a_2+ib_2) \longrightarrow \therefore \rho$ is antisymmetric.

③ Transitive:

Assume, $(a_1+ib_1)\,\rho\,(a_2+ib_2)$, $a_1 \le a_2 \wedge b_1 \le b_2$

& $(a_2+ib_2)\,\rho\,(a_3+ib_3)$, $a_2 \le a_3 \wedge b_2 \le b_3$

$\therefore a_1 \le a_3 \wedge b_1 \le b_3$

$\therefore (a_1+ib_1)\,\rho\,(a_3+ib_3)$ $\therefore \rho$ is Transitive.

$\therefore (C,\rho)$ is POSET. Proved.

2. $R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2)\}$

$A = \{1,2,3,4\}$

Prove R is an Equivalence Reln $\longrightarrow$ ① Reflexive ② Symmetric ③ Transitive.

① Reflexive

A relation on A is reflexive if $(a,a) \in R \ \forall a \in R$

Since $\{(1,1), (2,2), (3,3), (4,4)\} \in R$

$\therefore$ R is Reflexive.

② Symmetric

Relation is symmetric if $(a,b) \in R \Rightarrow (b,a) \in R$

Now, $(1,2) \in R \wedge (2,1) \in R$ ✓

$(1,3) \in R \wedge (3,1) \in R$ ✓

$(2,3) \in R \wedge (3,2) \in R$ ✓

$\therefore$ R is symmetric.

③ Transitive

Rule: $(a,b) \in R \wedge (b,c) \wedge R \Rightarrow (a,c) \in R$

Now, $(1,2) \in R \wedge (2,3) \in R \Rightarrow (1,3) \in R$

$(2,1) \in R \wedge (1,3) \in R \Rightarrow (2,3) \in R$

↳ satisfy for all cases.

$\therefore$ R is transitive.

Since all 3 conditions satisfied $\rightarrow$ R is an equivalence reln

3. $\rho = \{(a,b) : a, b \in \mathbb{Z} \wedge |a-b| \leq 5\}$ $\rightarrow$ Prove Equivalence.

① Reflexive: $\forall z \in \mathbb{Z} \rightarrow |a-a| = 0 \leq 5$

$\therefore (a,a) \in \rho$ $\therefore \rho$ is Reflexive.

② Symmetric : Rule: $(a,b) \in R \Rightarrow (b,a) \Rightarrow R$

Suppose, $(a,b) \in \rho \quad \therefore \ |a-b| \leq 5$

Since $|a-b| = |b-a|$, $|b-a| \leq 5$

$\therefore \ (b,a) \in \rho$

$\therefore \ \rho$ is symmetric.

③ Transitive : Rule: $(a,b) \in R \wedge (b,c) \in R \Rightarrow (a,c) \in R$

Suppose, $a=0, b=5, c=10$

$|a-b| = |0-5| = 5 \leq 5$ ✓

$|5-10| = 5 \leq 5$ ✓

$\therefore \ (0,5) \in \rho \wedge (5,10) \in \rho$

but, $|0-10| = 10 > 5$!!

$\therefore \ (0,10) \notin \rho$

$\therefore \ \rho$ is not transitive.

4. $f(x) = x^2+3x+1$, $g(x) = 2x-3$

$(f\circ g)(x) = f(g(x))$

$\therefore \ f(g(x)) = (2x-3)^2 + 3(2x-3) + 1$

$= 4x^2+9-12x+6x-9+1$

$\therefore \ (f\circ g)(x) = \boxed{4x^2-6x+1}$

$(g\circ f)(x) = g(f(x))$

$\therefore \ (g\circ f)(x) = 2(x^2+3x+1) - 3$

$= 2x^2+6x+2-3$

$(g\circ f)(x) = \boxed{2x^2+6x-1}$

5. $f : R \to R$, $f(x) = x^3$ is bijective (Prove). $f^{-1} = ?$ → inverse.

① one-one ② onto

① One one (injective)

let $f(a) = f(b)$

$\therefore a^3 = b^3$

$\Rightarrow a = b$

$\therefore f(a) = f(b)$

$\Rightarrow a = b$ $\therefore f$ is injective.

② Onto (Surjective) let, $y \in R$ (arbitary)

we need to find x such that $f(x) = y$ $(x \in R)$

$\because f(x) = x^3$

Choose $x = \sqrt[3]{y}$ $\therefore f(\sqrt[3]{y}) = (\sqrt[3]{y})^3 = y \in R$

thus every Real no. y has a pre-image in R.

$\therefore f$ is surjective.

$\therefore f$ is bijective (Proved)

$f^{-1} = ?$ let, $y = f(x) = x^3$

interchange x and y : $\boxed{x = y^3}$

$y = \sqrt[3]{x}$ $\therefore$ $\boxed{y^{-1}(x) = \sqrt[3]{x}}$ Ans.

7. Domain of $\sqrt{\frac{1-|x|}{2-|x|}} = ?$

$f(x) = \sqrt{\frac{1-|x|}{2-|x|}}$

For the sqrt to be defined,

$\frac{1-|x|}{2-|x|} \geq 0$

$\wedge$ $2-|x| \neq 0$

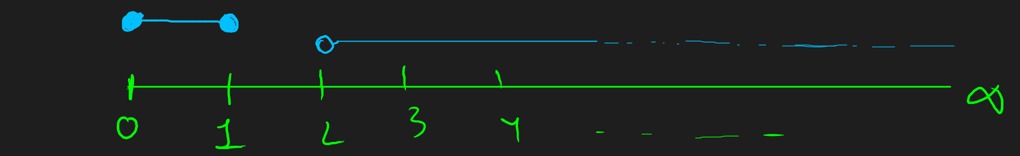
let, $t = |x|$

$\therefore \frac{1-t}{2-t} \geq 0$, $t \neq 2$

Critical points:

$1-t = 0 \Rightarrow t = 1$

$2-t = 0 \Rightarrow t = 2$



$t=1$ is allowed but $t=2$ is not allowed. & not $\in (1,2)$

$\therefore t \in [0,1] \cup (2,\infty)$.

$\therefore |x| \leq 1 \ \vee \ |x| > 2$

$\therefore$ Domain $= \{x : |x| \leq 1\} \cup \{x : |x| > 2\}$

8. Range of fn. $\log_2\left(\dfrac{\sin x - \cos x + 3\sqrt{2}}{\sqrt{2}}\right)$

$\sin x - \cos x = \sqrt{2}\sin(x-\alpha)$

Now, $\sin(x-\alpha) = \sin x \cos\alpha - \cos x \sin\alpha$

$\therefore \sqrt{2}\sin(x-\alpha) = \sqrt{2}\sin x \cos\alpha - \sqrt{2}\cos x \sin\alpha$

$\therefore \sin x - \cos x = (\sqrt{2}\cos\alpha)\sin x - (\sqrt{2}\sin\alpha)\cos x$

$a\sin x + b\cos x = R\sin(x+\alpha)$

where $R = \sqrt{a^2+b^2}$

here, $a=1, b=1$

$R = \sqrt{1^2+1^2} = \sqrt{2}$

$\therefore \sqrt{2}\cos\alpha = 1$, $\sqrt{2}\sin\alpha = 1$

$\alpha = \cos^{-1}\left(\dfrac{1}{\sqrt{2}}\right)$, $\alpha = \sin^{-1}\left(\dfrac{1}{\sqrt{2}}\right) = \dfrac{\pi}{4}$

$= \dfrac{\pi}{4}$

$\therefore \sin x - \cos x = \sqrt{2}\sin(x - \pi/4)$

$\therefore R(\sin x - \cos x) = [-\sqrt{2}, \sqrt{2}]$

$\sin x - \cos x$

$= \sin x - \sin(\pi/2 - x)$

Now, let $A = \dfrac{\sin x - \cos x + 3\sqrt{2}}{\sqrt{2}}$

Since $-\sqrt{2} \leq \sin x - \cos x \leq \sqrt{2}$

$\Rightarrow -\sqrt{2}+3\sqrt{2} \leq \sin x - \cos x + 3\sqrt{2} \leq \sqrt{2}+3\sqrt{2}$ (adding $3\sqrt{2}$)

$\Rightarrow \dfrac{-\sqrt{2}+3\sqrt{2}}{\sqrt{2}} \leq A \leq \dfrac{\sqrt{2}+3\sqrt{2}}{\sqrt{2}}$ (dividing by $\sqrt{2}$)

$\Rightarrow \quad 2 \leq A \leq 4$

$\Rightarrow \quad \log_2(2) \leq \log_2(A) \leq \log_2(4)$ [Apply $\log_2$]

$\Rightarrow \quad 1 \leq \log_2\left(\frac{\sin x - \cos x + 3\sqrt{2}}{\sqrt{2}}\right) \leq 2$

$\therefore \quad R(f(x)) = [1, 2]$ Ans

9. Range of $\frac{x}{x^2+1} = ?$

$f(x) = \frac{x}{x^2+1}$

We know $\forall x \in \mathbb{R}$

$(x-1)^2 \geq 0$

$\Rightarrow x^2 - 2x + 1 \geq 0$

$\Rightarrow x^2 + 1 \geq 2x \quad \therefore \boxed{\frac{x}{x^2+1} \leq \frac{1}{2}}$

and, $(x+1)^2 \geq 0$

$\Rightarrow x^2 + 2x + 1 \geq 0$

$\Rightarrow x^2 + 1 \geq -2x$

$\Rightarrow -(x^2+1) \leq 2x$

$\Rightarrow -\frac{1}{2} \leq \frac{x}{x^2+1}$

$\therefore \boxed{\frac{x}{x^2+1} \geq -\frac{1}{2}}$

$\therefore \boxed{-\frac{1}{2} \leq \frac{x}{x^2+1} \leq \frac{1}{2}}$ Ans

10. Hasse diagram of the Poset $(A, /)$ where $A = \{1,2,3,4,5,6,7,8\}$

$A = \{1,2,3,4,5,6,7,8\}$

A cover relⁿ $a \prec b$ exists when: $a \mid b$ and there is no element c such that $a \mid b \mid c$

$1 \mid 2, \; 2 \mid 4 \longrightarrow$ intermediate elmt 2

$\therefore$ we cannot connect 1 to 4 directly.

Divisibility Relⁿ

- 1 divides everything
- 2 divides 2, 4, 6, 8
- 3 divides 6
- 4 divides 8

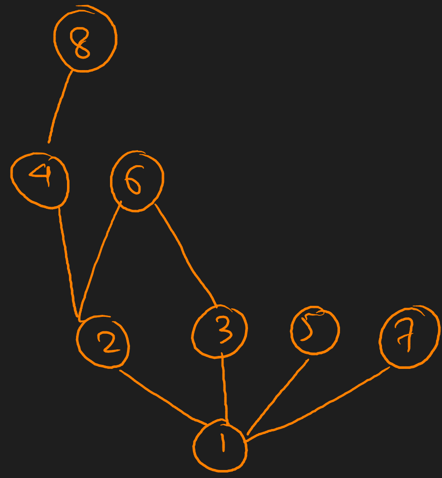
Keep only immediate divisibility

$1 \prec 2, \quad 1 \prec 3, \quad 1 \prec 5, \quad 1 \prec 7$

$2 \prec 4, \quad 2 \prec 6$

$3 \prec 6$

$4 \prec 8$



11. $S = \{1, 2, 3, 5, 6, 10, 15, 30\}$

Cover relations:

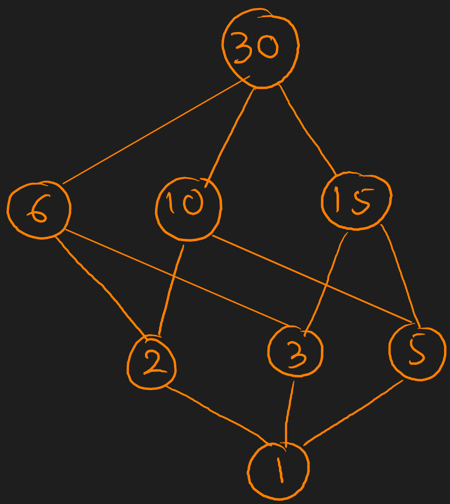
$1 \prec 2, \quad 1 \prec 3, \quad 1 \prec 5,$

$2 \prec 6, \quad 2 \prec 10$

$3 \prec 6, \quad 3 \prec 15$

$5 \prec 10, \quad 5 \prec 15$

$6 \prec 30, \quad 10 \prec 30, \quad 15 \prec 30$



12. Poset (D_{40}, P) where D_{40} = set of all +ve divisors of 40 and P is relation division.

$40 = 2^3 \times 5$

$D_{40} = \{1, 2, 4, 5, 8, 10, 20, 40\}$

Cover Relation:

$1 \prec 2, \quad 1 \prec 5$

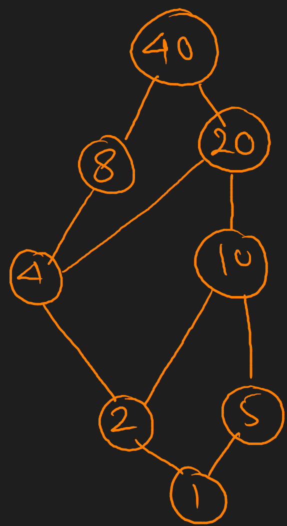
$2 \prec 4, \quad 2 \prec 10$

$4 \prec 8, \quad 4 \prec 20$

$8 \prec 40, \quad 5 \prec 10$

$10 \prec 20$

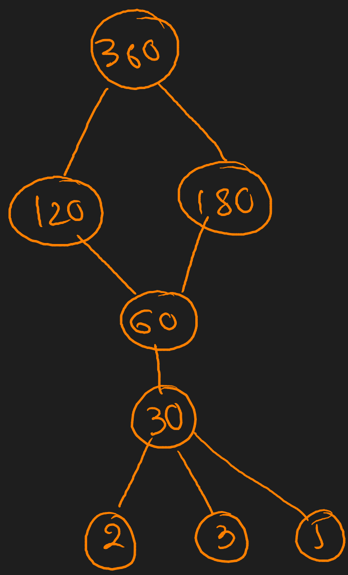
$20 \prec 40$



13. $S = \{2, 3, 5, 30, 60, 120, 180, 360\}$

Cover Relations

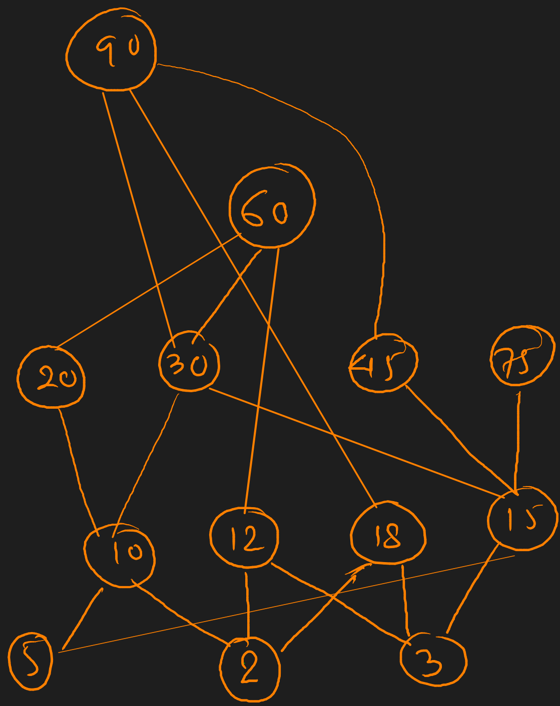
$2 \prec 30$,
$3 \prec 30$,
$5 \prec 30$,
$30 \prec 60$,
$60 \prec 120$, $60 \prec 180$
$120 \prec 360$
$180 \prec 360$.



14. (A, R) where $A = \{2, 3, 5, 10, 12, 15, 18, 20, 30, 45, 60, 75, 90)$

Cover Relations.

$2 \prec 10$, $2 \prec 12$, $2 \prec 18$
$3 \prec 12$, $3 \prec 15$, $3 \prec 18$,
$5 \prec 10$, $5 \prec 15$
$10 \prec 20$, $10 \prec 30$
$12 \prec 60$
$15 \prec 30$, $15 \prec 45$, $15 \prec 75$
$18 \prec 90$
$20 \prec 60$
$30 \prec 60$, $30 \prec 90$
$45 \prec 90$



15. let, $d = \gcd(2a+b, a+2b)$

we have to prove that $d = 1$ or $d = 3$

given $\gcd(a, b) = 1$

since $d = \gcd(2a+b, a+2b)$

$\therefore d \mid (2a+b)$ and $d \mid (a+2b)$ $\longrightarrow$ d devides both.

Now, $2(a+2b) - (2a+b) = 3b$

$\therefore d \mid 3b$

Similarly, $2(2a+b) - (a+2b) = 3a$

$\therefore d \mid 3a$.

Theorem

if $d \mid x \wedge d \mid y$,
then d devides any linear comb^n
$mx + ny$

$\therefore d \mid 3a \wedge d \mid 3b$

$\therefore d \mid \gcd(3a, 3b)$

Since $\gcd(a,b) = 1$, $\gcd(3a, 3b) = 3$ $\therefore \boxed{d \mid 3}$

Now, the +ve dividors of 3 are only 1 & 3

$\therefore \gcd(2a+b, a+2b) = 1$ or 3 (Proved)

16. given $\gcd(a,4) = 2$ and $\gcd(b,4) = 2$ prove. $\gcd(a+b, 4) = 4$

$\underline{\gcd(a,4) = 2} \longrightarrow$

- a is divisible by 2
- But a is not divisible bey 4.

$\therefore a = 4m + 2$, $m \in \mathbb{Z}$

& similarly $b = 4n + 2$, $n \in \mathbb{Z}$

Now, $a + b = (4m+2) + (4n+2) = 4(m+n) + 4$
$= 4(m+n+1)$

$\therefore 4 \mid (a+b)$

since, $4 \mid (a+b)$ and obviously $4 \mid 4$

$\boxed{\gcd(a+b, 4) = 4}$ Proved

17. $d = \gcd(a+b, a-b)$ prove $d = 1$ or 2

given $\gcd(a,b) = 1$

since $d = \gcd(a+b, a-b)$

$\therefore d \mid (a+b) \wedge d \mid (a-b)$

Now, $(a+b) + (a-b) = 2a$ $\therefore d \mid 2a$

& $(a+b) - (a-b) = 2b$ $\therefore d \mid 2b$.

$\therefore d \mid 2a \wedge d \mid 2b$

$\therefore d = \gcd(2a, 2b)$

$\because \gcd(a,b) = 1$

$\therefore d = \gcd(2a, 2b) = 2$.

Now, the +ve divisors of 2 are only 1 and 2

$\therefore$ $\gcd(a+b, a-b) = 2$ or 1 (Proved)

18.

if n is odd $n^2 = 8k+1$ $\longrightarrow$ Prove.
for some $k \in \mathbb{Z}^+$

let, $n \in \mathbb{Z} \wedge n = \text{odd}$

$\therefore n = 2m+1, \; m \in \mathbb{Z}$

$\Rightarrow n^2 = (2m+1)^2 = 4m^2+1+4m = 4m(m+1)+1$

Since m and $(m+1)$ are consecutive integers one of them is even.

$\therefore m(m+1) = 2k$

$\therefore n^2 = 4(2k)+1$ $\quad \therefore \boxed{n^2 = 8k+1}$ for some $k \in \mathbb{Z}^+$

Therefore, the sq. of every odd integer of the form $8k+1$ ✓

19. Prove, every perfect square is of the form $9k, 3k+1$,

Any integer n written in 3 forms (any one) $\quad k \in \mathbb{Z}^+$

$n = 3m \;, \; n = 3m+1, \; n = 3m+2$

Case 1: $\boxed{3m}$: $n^2 = (3m)^2 = 9m^2$ $\quad \therefore n^2 = 9k$

Case 2: $\boxed{3m+1}$: $n^2 = (3m+1)^2 = 9m^2+1+6m = 3(\underline{3m^2+2m})+1$

$\therefore n^2 = 3k+1$

Case 3 $\boxed{3m+2}$: $n^2 = (3m+2)^2 = 9m^2+4+6m = 3(\underline{3m^2+2m+1})+1$

$\therefore n^2 = 3k+1$

$\therefore$ in all cases $n^2 = 9k$ or $n^2 = 3k+1$

$\therefore$ Proved

20. Prove: $n^3 = 9k, 9k\pm 1$ $\forall n \in \mathbb{Z}$ $\vee$ same $k \in \mathbb{Z}$

any integer n is in one of the forms —

$3m, 3m+1, 3m-1$

$\boxed{3m}$:

$n = 3m$

$\Rightarrow n^3 = (3m)^3$

$\Rightarrow n^3 = 27m^3$

$= 9.(3m^3)$

$\therefore \boxed{n^3 = 9k}$

$\boxed{3m+1}$:

$n = 3m+1$

$\Rightarrow n^3 = (3m+1)^3$

$\Rightarrow n^3 = 27m^3 + 27m + 9m + 1$

$= 9(3m^3 + 3m + m) + 1$

$\therefore \boxed{n^3 = 9k+1}$

$\boxed{3m-1}$: $n = 3m-1$

$\Rightarrow n^3 = (3m-1)^3$

$= 27m^3 - 27m + 9m - 1$

$= 9(3m^3 + 3m + m) + 1$

$\therefore \boxed{n^3 = 9k+1}$

21. Proof: Any 3 consecutive integers product divisible by 3.

let, $n, (n+1), (n+2)$

any integer n is one of the forms $3k, 3k+1, 3k+2$

$\boxed{3k}$:

$n(n+1)(n+2)$

$= 3k(3k+1)(3k+2)$

$= (9k^2+3k)(3k+2)$

$= 27k^3 + 18k^2 + 9k^2 + 6k$

$= 27k^3 + 27k^2 + 6k$

$= 3(9k^3 + 9k^2 + 2k)$

this is divisible by 3.

$\boxed{3k+1}$

$n(n+1)(n+2)$

$= (3k+1)(3k+2)(3k+3)$

$= (9k^2 + 9k + 2)(3k+3)$

$= 27k^3 + 27k^2 + 27k^2 + 27k + 6$

$= 27k^3 + 54k^2 + 27k + 6$

$= 3(9k^3 + 18k^2 + 9k + 2)$

divisible by 3.

$\boxed{3k+2}$

$n(n+1)(n+2)$

$= (3k+2)(3k+3)(3k+4)$

$= (9k^2 + 15k + 6)(3k+4)$

$= 27k^3 + 36k^2 + 45k^2 + 60k$
$+ 18k + 24$

$= 27k^3 + 81k^2 + 78k + 24$

$= 3(9k^3 + 27k^2 + 26k + 8)$

divisible by 3

$\therefore 3 \mid n(n+1)(n+2)$ $\forall n \in \mathbb{Z}$ proved

22. Prove. Product of any r consecutive natural numbers are divisible by r.

Prove $r! \mid (n+1)(n+2)\cdots(n+r)$

Since binomial coefficients are integers $^{n+r}C_r \in \mathbb{Z}$

But. $^{n+r}C_r = \dfrac{(n+r)!}{n!\,r!} = \dfrac{(n+1)(n+2)\cdots(n+r)}{r!}$

Since the $^{n+r}C_r \in \mathbb{Z}$

$$\therefore \frac{(n+1)(n+2)\cdots(n+r)}{r!} \in \mathbb{Z}$$

$$\therefore r! \mid (n+1)(n-2)\cdots(n+r) \quad \text{(Proved)}$$

23. All solns in the integers of $5x+3y=52$

$5x+3y=52$

$\therefore 3y = 52-5x$

$$\boxed{y = \frac{52-5x}{3}}$$

for y to be an integer

$(52-5x) \equiv 0 \pmod 3$

$52 \equiv 1 \pmod 3$

$5 \equiv 2 \pmod 3$

Substitute

$52 \to 1$

$5 \to 2$

$\therefore (1-2x) \equiv 0 \pmod 3$

$2x \equiv 1 \pmod 3$

Check inverse of 2 modulo 3.

$2 \times 2 = 4 \equiv 1 \pmod 3$

$\therefore$ the inverse of 2 is 2.

$\therefore$ Multiply both sides with 2

$2(2x) \equiv 2 \pmod 3$

$4x \equiv 2 \pmod 3$

Since $4 \equiv 1 \pmod 3$

$\therefore x \equiv 2 \pmod 3$

$\therefore x = 3k+2 \longrightarrow \therefore$ possible values of $\boxed{x = 2, 5, 8}$

Now, $5x+3y=52$

& $\because y \geq 1$

$\therefore 5x < 52$

$x < (52/5)$

$x < 10.4$

$x = 3(0)+2 = 2$ ✓

$x = 3(1)+2 = 5$ ✓

$x = 3(2)+2 = 8$ ✓

$x = 3(3)+2 = 11$ ✗

$\therefore y = \dfrac{52-5x}{3}$

$$\boxed{\begin{aligned} y(2) &= \frac{52-5(2)}{3} = 14 \\ y(5) &= \frac{52-5(5)}{3} = 9 \\ y(8) &= \frac{52-5(8)}{3} = 4 \end{aligned}}$$

Ans

24. Solve diophantine eqn. $158x - 47y = 9$

$\gcd(158, 47)$

use euclid's algo -

$158 = 47 \times 3 + 17$
$47 = 17 \times 2 + 13$
$17 = 13 \times 1 + 4$
$13 = 4 \times 3 + 1$
$4 = 1 \times 4 + 0$

$\therefore \gcd(158, 47) = 1$

here, $1 | 9$

Express 1 as linear comb.

$1 = 13 - 4 \times 3$
$= 13 - (17 - 13) \times 3$
$= 13 - 17 \times 3 + 13 \times 3$
$= 4 \times 13 - 3 \times 17$
$= 4 \times (47 - 2 \times 17) - 3 \times 17$
$= 4 \times 47 - 8 \times 17 - 3 \times 17$
$= 4 \times 47 - 11 \times 17$
$= 4 \times 47 - 11 \times (158 - 47 \times 3)$
$= 4 \times 47 - 11 \times 158 + 33 \times 47$
$= 37 \times 47 - 11 \times 158$

$\therefore 9 = (-99) \times \boxed{158} - (-333) \times \boxed{47}$

$\therefore x_0 = -99$ $y_0 = -333$

general soln: $ax + by = c \Longrightarrow d = \gcd(a, b) = 1$

$\boxed{x = x_0 + \frac{b}{d} t}$ $\boxed{y = y_0 - \frac{a}{d} t}$

here, $a = 158$, $b = -47$, $d = 1$

$\therefore$ $x = -99 - 47t$
$y = -333 - 158t$ $\Big|$ $t \in \mathbb{Z}$ Ans.

25. Find all +ve int solns. $12x + 201y = 1$.

$\gcd(12, 201)$

use euclid's algo

$201 = 12 \times 16 + 9$
$12 = 9 \times 1 + 3$
$9 = 3 \times 3 + 0$

$\therefore \gcd(12, 201) = 3$

here, $3 \nmid 1$

For a Diophantine eqn $ax + by = c$ to have integer solutions we must have $\gcd(a, b) | c$

but here $3 \nmid 1$

$\therefore$ the eqn has no integer solns. Ans.

26. Find sol^ns of the eq^n: $243x+198y=9$

gcd(243, 198)

Euclid's algo

$243 = 198\times1+45$

$198 = 45\times4+18$

$45 = 18\times2+9$

$18 = 9\times2+0$

$\therefore$ gcd(243, 198) = 9

here $9|9$

soln exist.

Express 9 as linear combinatn

$9 = 45-18\times2$

$= 45-(198-45\times4)\times2$

$= 45-198\times2+45\times8$

$= 9\times45-2\times198$

$= 9\times(243-198)-2\times198$

$= 9\times243-9\times198-2\times198$

$= 9\times243-11\times198$

$= 243(9)+198(-11)$

$\boxed{x_0=9}$, $\boxed{y_0=-11}$

general soln for $ax+by+c$, $d=\gcd(a,b)=1$

$\boxed{x=x_0+\frac{b}{d}t}$ $\boxed{y=y_0-\frac{a}{d}t}$

here, $a=243$, $b=198$, $d=9$

$\therefore x = 9+\frac{198}{9}t = 9+22t$

$y=-11-\frac{243}{9}t = -11-27t$

$\boxed{\begin{array}{l}\therefore x=22t+9 \\ \therefore y=-27t-11\end{array}}$ Ans

27. Prove $3^{302}\equiv4 \pmod 5$

$3^1\equiv3 \pmod 5$

$3^2\equiv4 \pmod 5$

$3^3\equiv2 \pmod 5$

$3^4\equiv1 \pmod 5$

$3^5\equiv3 \pmod 5$

Repeat every 4

3, 4, 2, 1, 3, 4, 2, 1 .....

Now, $302 = 4\times75+2$

$\therefore 302\equiv2 \pmod 4$

powers of 3 modulo 5 repeats every 4

$\therefore 3^{302}\equiv3^2 \pmod 5$

here, $3^2=9\equiv4 \pmod 5$

$\therefore 3^{302}\equiv4 \pmod 5$

(Proved)

28. Prove $11^{34} \equiv 2 \pmod{17}$

Fermat Theorem

if p is prime & $\gcd(a,p)=1$, then $\boxed{a^{p-1} \equiv 1 \pmod{p}}$

using Fermat's Little theorem –

Since 17 is prime and $\gcd(11,17)=1$

$\therefore\ 11^{16} \equiv 1 \pmod{17}$

Now, $34 = 16 \times 2 + 2$

$\therefore\ 11^{34} = 11^{(16\times 2+2)} = (11^{16})^2 \cdot 11^2$

taking modulo 17 $\Rightarrow$ $11^{34} \equiv (11^{16})^2 \cdot 11^2 \pmod{17}$

$\Rightarrow\ 11^{34} \equiv (1)^2 \cdot 11^2 \pmod{17}$

$\Rightarrow\ 11^{34} \equiv 11^2 \pmod{17}$

Now, $11^2 = 121 = 7 \times 17 + 2$

$\therefore\ 11^2 \equiv 2 \pmod{17}$

$\therefore\ \boxed{11^{34} \equiv 2 \pmod{17}}$

Proved

29. $7^{32} \equiv ? \pmod{5}$

$7 \equiv 2 \pmod{5}$

$\Rightarrow\ 7^{32} \equiv 2^{32} \pmod{5}$

Now, $2^4 \equiv 1 \pmod{5}$

$\therefore\ 2^{32} = (2^4)^8 \equiv (1)^8 \pmod{5}$

$\therefore\ 7^{32} \equiv 1 \pmod{5}$

Ans.

30. unit digit in 3^{909}.

$3^1 = 3$ (unit digit = 3)

$3^2 = 9$ (unit digit = 9)

$3^3 = 27$ (unit digit = 7)

$3^4 = 81$ (unit digit = 1)

$3^5 = 243$ (unit digit = 3)

repeats after every 4

cycle length = 4 $\quad (3, 9, 7, 1)$

Now, $909 = 4 \times 227 + 1$

$\therefore\ 909 \equiv 1 \pmod{4}$ $\rightarrow$ 1 means we take the 1st term of the cycle.

$\therefore$ unit digit in $3^{909} = 3$

31. unit digit in 13^{37}

$13^1 = 13 \rightarrow ud = 3$
$13^2 = 169 \rightarrow ud = 9$
$13^3 = 2197 \rightarrow ud = 7$
$13^4 = 28561 \rightarrow ud = 1$
$13^5 = 371293 \rightarrow ud = 3$
$13^6 = 4826809 \rightarrow ud = 9$

cycle length = 4 $(3, 9, 7, 1)$

Now, $37 = 4 \times 9 + 1$

$\therefore 37 \equiv 1 \pmod 4$

$\therefore$ unit digit in $13^{37} = 3$

32. $2^{18} + 9^{18} \equiv 0 \pmod{17} \rightarrow$ Prove

using fermat's theorm for prime 17

$a^{16} \equiv 1 \pmod{17}$

here, $2^{18} = 2^{16} \cdot 2^2 \equiv 1 \cdot 4 \pmod{17}$ $[\because a^{16} \equiv 1 \pmod{17}$ $\therefore 2^{16} \equiv 1 \pmod{17}]$

$9^{18} = 9^{16} \cdot 9^2 \equiv 1 \cdot 13 \pmod{17}$ and $4 \equiv 4 \pmod{17}$

$[\because 9^{16} \equiv 1 \pmod{17}$ & $81 \equiv 13 \pmod{17}]$

$\therefore (2^{18} + 9^{18}) = 4 + 13 = 17 \equiv 0 \pmod{17}$ Proved

33. Solve linear congruence $6x \equiv 3 \pmod 9$

$\gcd(6, 9) = 3 \rightarrow$ since $3 | 3$ solution exist

$6x \equiv 3 \pmod 9$

$\Rightarrow 2x \equiv 1 \pmod 3$

multiply both side by 2

$\Rightarrow 2(2x) \equiv 2(1) \pmod 3$

$\Rightarrow 4x \equiv 2 \pmod 3$

Since $4 \equiv 1 \pmod 3$ $\therefore$ $\boxed{x \equiv 2 \pmod 3}$

inverse of 2 mod 3

$2 \cdot 2 = 4 \equiv 1 \pmod 3$

$\therefore$ the inverse of 2 mod 3 is $\boxed{2}$

$\therefore\ x = 3k+2$ for some int k

Now. $k=0 \Rightarrow x=2$

$k=1 \Rightarrow x=5$

$k=2 \Rightarrow x=8$

$k=3 \Rightarrow x=11$

Ans: $\boxed{2,5,8}$

34. linear congruence $9x \equiv 21 \pmod{30}$

$\gcd(9,30) = 3 \rightarrow$ since $3|21$ solutⁿ exist.

$9x \equiv 21 \pmod{30}$

$\Rightarrow 3x \equiv 7 \pmod{10}$ $\times 7$

$\Rightarrow 7(3x) \equiv 7(7) \pmod{10}$

$\Rightarrow 21x \equiv 49 \pmod{10}$

$\Rightarrow x \equiv 9 \pmod{10}$

$\therefore\ x = 10k+9$

inverse of 3 mod 10

$3.7 = 21 \equiv 1 \pmod{10}$

$\therefore$ the inverse of 3 mod 10 = 7

Now, $k=0 \Rightarrow x=9$

$k=1 \Rightarrow x = 10+9 = 19$

$k=2 \Rightarrow x = 20+9 = 29$

$k=3 \Rightarrow x = 30+9 = 39$

Ans. $x = \boxed{9, 19, 29} \pmod{30}$

35. linear congruence $18x \equiv 30 \pmod{42}$

$\gcd(18,42) = 6$, since $6|30 \rightarrow$ Solⁿ exist.

$\therefore\ 3x \equiv 5 \pmod 7$ $\times 5$

$\Rightarrow 5(3x) \equiv 5(5) \pmod 7$

$\Rightarrow 15x \equiv 25 \pmod 7$

$\Rightarrow x \equiv 4 \pmod 7$

$\therefore\ x = 7k+4$

inverse of 3 mod 7

$3 \times 5 = 15 \equiv 1 \pmod 7$

$\therefore$ inverse $= 5$

Now, $k=0 \Rightarrow x=4$

$k=1 \Rightarrow x=11$

$k=2 \Rightarrow x=18$

$k=3 \Rightarrow x=25$

$k=4 \Rightarrow x=32$

$k=5 \Rightarrow x=39$

$k=6 \Rightarrow x=46$

$\therefore$ Ans $x = 4, 11, 18, 25, 32, 39 \pmod{42}$

36. Linear congruence $345x \equiv 18 \pmod{912}$

$\gcd(345, 912)$

apply euclid's algo

$912 = 345 \times 2 + 222$

$345 = 222 \times 1 + 123$

$222 = 123 \times 1 + 99$

$123 = 99 \times 1 + 24$

$99 = 24 \times 4 + 3$

$24 = 3 \times 8 + 0$

$\therefore \gcd(345, 912) = 3$

Since $3 \mid 912$

$\therefore$ Solution exist.

$\therefore \left(\frac{345}{3}\right) x \equiv \left(\frac{18}{3}\right) \pmod{304}$

$\Rightarrow 115x \equiv 6 \pmod{304}$

multiply with (-37)

$\Rightarrow 115(-37)x \equiv 6(-37) \pmod{304}$

$\Rightarrow x \equiv -222 \pmod{304}$

- inverse of 115 mod 304

Use Back Substitute

$3 = 99 - 24 \times 4$

$= 99 - (123 - 99)4$

$= 99 - 123 \times 4 + 99 \times 4$

$= 5 \times 99 - 123 \times 4$

$= 5(222 - 123) - 123 \times 4$

$= 5 \times 222 - 123 \times 5 - 123 \times 4$

$= 5 \times 222 - 123 \times 9$

$= 5 \times 222 - (345 - 222) \times 9$

$= 5 \times 222 - 345 \times 9 + 222 \times 9$

$= 14 \times 222 - 345 \times 9$

$= 14(912 - 345 \times 2) - 345 \times 9$

$= 14 \times 912 - 28 \times 345 - 9 \times 345$

$3 = 14 \times 912 - 37 \times 345$

$\therefore 1 = 14 \times 304 - 37 \times 115$

Now take both side (mod 304)

$1 \equiv 0 + 115(-37) \pmod{304}$

$115(-37) \equiv 1 \pmod{304}$

So the inverse of 115 modulo 304 is (-37)

$\therefore x = 304k - 222$

Now,

$k = 0 \Rightarrow x = -222$

$k = 1 \Rightarrow x = 82$

$k = 2 \Rightarrow x = 386$

$k = 3 \Rightarrow x = 690$

$k = 4 \Rightarrow x = 994$

Ans:

$\boxed{x \equiv 82, 386, 690 \pmod{912}}$

37. Inverse of 19 mod 141

$19x \equiv 1 \pmod{141}$

gcd (19,141)

apply euclid's algo

$141 = 19 \times 7 + 8$
$19 = 8 \times 2 + 3$
$8 = 3 \times 2 + 2$
$3 = 2 \times 1 + 1$
$2 = 1 \times 2 + 0$

$\therefore \gcd(19,141) = 1$

$19^{-1} \pmod{141} = ?$

Now, backsubstitute :—

$1 = 3 - 2$
$= 3 - (8 - 3 \times 2)$
$= 3 \times 3 - 8$
$= 3(19 - 8 \times 2) - 8$
$= 3 \times 19 - 8 \times 6 - 8$
$= 3 \times 19 - 8 \times 7$
$= 3 \times 19 - (141 - 19 \times 7) \times 7$
$= 3 \times 19 - 141 \times 7 + 19 \times 7 \times 7$
$1 = 52 \times 19 - 7 \times 141$

$\therefore 1 \equiv 52 \times 19 - 0 \pmod{141}$

or, $52 \times 19 \equiv 1 \pmod{141}$

$\therefore 19^{-1} \pmod{141} = 52$ Ans.

38. $144^{-1} \pmod{233} = ?$

$144x \equiv 1 \pmod{233}$

gcd (144, 233)

use euclid's algo,

$233 = 144 \times 1 + 89$
$144 = 89 \times 1 + 55$
$89 = 55 \times 1 + 34$
$55 = 34 \times 1 + 21$
$34 = 21 \times 1 + 13$
$21 = 13 \times 1 + 8$
$13 = 8 \times 1 + 5$
$8 = 5 \times 1 + 3$
$5 = 3 \times 1 + 2$
$3 = 2 \times 1 + 1$
$2 = 1 \times 2 + 0$

$\therefore \gcd(144, 233) = 1$

Now, backsubstitution —

$1 = 3 - 2$
$= 3 - (5 - 3)$
$= 3 \times 2 - 5$
$= (8 - 5) \times 2 - 5$
$= 8 \times 2 - 5 \times 2 - 5$
$= 8 \times 2 - 3 \times 5$
$= 8 \times 2 - 3(13 - 8)$
$= 8 \times 2 - 13 \times 3 + 8 \times 3$
$= 8 \times 5 - 13 \times 3$
$= (21 - 13) \times 5 - 13 \times 3$
$= 21 \times 5 - 13 \times 5 - 13 \times 3$
$= 21 \times 5 - 8 \times 13$
$= 5 \times 21 - 8 \times (34 - 21)$
$= 5 \times 21 - 8 \times 34 + 8 \times 21$
$= 13 \times 21 - 8 \times 34$
$= 13 \times (55 - 34) - 8 \times 34$

$= 13\times(55-34) - 8\times 34$
$= 13\times 55 - 13\times 34 - 8\times 34$
$= 13\times 55 - 21\times 34$
$= 13\times 55 - 21\,(89-55)$
$= 13\times 55 - 21\times 89 + 21\times 55$
$= 34\times 55 - 21\times 89$
$= 34\times(144-89) - 21\times 89$
$= 34\times 144 - 34\times 89 - 21\times 89$
$= 34\times 144 - 55\times 89$
$= 34\times 144 - 55\,(233-144)$
$= 34\times 144 - 55\times 233 + 55\times 144$
$= 89\times 144 - 55\times 233$

$\therefore\ 1 = 89\times 144 - 55\times 233$

$\therefore\ 1 \equiv 89\times 144 \pmod{233}$

Or, $89\times 144 \equiv 1 \pmod{233}$

$\therefore\ 144^{-1} \pmod{233} = \boxed{89}$

Ans.

39. Find all integers $m \geq 5$ such that $7 \equiv m^2 \pmod m$

$m^2 \equiv 0 \pmod m$ b/c $m \mid m^2$

$\therefore\ 7 \equiv 0 \pmod m$

means $m \mid 7$

Since 7 is prime its +ve divisors are 1, 7.

given $m \geq 5$ $\therefore$ the only possibility is $\boxed{m = 7}$

Ans

40. $n = a^2+b^2\ \forall a,b \in \mathbb{Z}$ then $n = 4q$ or $4q+1$ or $4q+2$ for some $q \in \mathbb{Z}$

Hence deduce the 123456 cannot be written as sum of 2 sq.

any int is either $(2k)$ or $(2k+1)$

$\boxed{2k}$: $a^2 = (2k)^2 = 4k^2 \equiv 0 \pmod 4$

$\boxed{2k+1}$: $a^2 = (2k+1)^2 = 4k^2+1+4k = 4(k^2+k)+1 \equiv 1 \pmod 4$

Hence, every square is congruent to either 0 or 1 $\pmod 4$.

Same for b^2.

possible values of $a^2+b^2 \pmod 4 = \begin{cases} 0+0=0, \\ 0+1=1, \\ 1+0=1, \\ 1+1=2 \end{cases} \pmod 4$

$\therefore n = a^2+b^2$ can be congruent to $0, 1, 2 \pmod 4$

$\therefore n = 4q \vee n = 4q+1 \vee n = 4q+2$ for some $q \in \mathbb{Z}$.

<u>1234567</u>

Find remainder mod 4.

$1234567 = 4(308641) + 3$

$\therefore 1234567 \equiv 3 \pmod 4$

but a sum of 2 sq. can't be congruent to $3 \pmod 4$.

$\therefore 1234567$ can't be written as sum of 2 sq.